

Fossil gems

Ammonite fossils can be as beautiful as jewels. Some look like glass ornaments when they are cut open and polished, their once-hollow shells having filled up with crystalline minerals over millions of years. Others have a pearly surface that shimmers with color, forming one of the world's most precious gemstones.

Rare jewels

In 1981, the World Jewellery Confederation gave official gemstone status to a brightly colored mineral found only on the surface of certain ammonite fossils. It is thought to be one of the rarest gemstones on the Earth, rivaling red diamond. It is found only in a few parts of the Rocky Mountains in North America and used to make exclusive luxury jewelry.



▼ AMMONITES GREW in a spiral shape, adding new chambers to their shell as they got bigger. This fossil of the ammonite *Desmoceras* is about 100 million years old.



PEARLY AMMONITES

Ammonites made their shells from the mineral aragonite – the shiny mineral from which pearls form. In most fossils the shell has entirely disappeared and all that remains is a mould of its hollow interior. However, some ammonite fossils retain a pearly film of aragonite on the surface. In the best specimens, this delicate layer produces shimmering colours by splitting reflected light, a phenomenon known as iridescence.

